Brain Transplantation: Scientific Feasibility, Neurobiological Barriers, Ethical Constraints, and a Responsible Research Roadmap

A Critical Narrative Review and Research Agenda

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Scientific status statement

No human whole-brain transplant or whole-body/cephalosomatic transplant with functional spinal-cord reconnection has been demonstrated. This manuscript does not report original experiments, does not propose a clinical operation, and does not treat cellular recovery as evidence of restored consciousness or personhood.

Abstract

Brain transplantation occupies an unusual boundary between transplant surgery, neurocritical care, spinal-cord repair, regenerative medicine, and the philosophy of personal identity. Public discussion often combines two distinct concepts: transplantation of an isolated donor brain into a recipient body, and transplantation of a recipient head onto a donor body. Neither procedure is clinically available. This critical narrative review evaluates the scientific evidence relevant to feasibility, identifies the interfaces that would require successful reconstruction, distinguishes cellular preservation from integrated neurological recovery, and proposes a responsible research agenda. Historical animal cephalic-exchange experiments established that a transplanted head could receive short-term blood flow, but they did not reconnect the spinal cord and therefore did not restore bodily motor or sensory integration. More recent BrainEx and OrganEx studies demonstrated that selected microcirculatory, metabolic, molecular, and cellular functions can be preserved or restored after prolonged ischaemia in pigs; importantly, these experiments did not demonstrate global organized brain activity, consciousness, or transplantability. The decisive barriers are not a single vascular anastomosis but simultaneous control of cerebral ischaemia-reperfusion injury, reconstruction of complete cervical spinal pathways, integration of cranial and autonomic systems, maintenance of cerebrospinal-fluid and endocrine physiology, prevention of alloimmune injury, and resolution of consent, identity, death determination, justice, and animal-welfare concerns. Current spinal-cord regeneration and neuromodulation research provides valuable component knowledge but remains far from the reproducible, high-bandwidth reconnection required after complete cervical transection. We conclude that a human brain or whole-body transplant is scientifically unvalidated and ethically unjustifiable at present. Legitimate progress should focus on lower-risk component research with predefined functional endpoints, independent oversight, transparent reporting, and explicit stop rules. The appropriate near-term objective is not a human transplant attempt, but rigorous solutions to organ preservation, neural-circuit repair, immune tolerance, and neuroethical governance.

Keywords: brain transplantation; head transplantation; cephalosomatic transplantation; spinal cord injury; neuroregeneration; organ preservation; BrainEx; OrganEx; neuroethics; personal identity

Key Points

- “Brain transplant” and “head/body transplant” are anatomically and ethically different procedures and should not be treated as synonyms.
- Cellular recovery after ischaemia is not equivalent to restoration of integrated brain function, consciousness, memory, or personal identity.
- Complete cervical spinal-cord reconnection is the principal unsolved functional barrier for whole-body transplantation; isolated whole-brain transplantation adds cranial-nerve, pituitary, meningeal, venous-sinus, and cerebrospinal-fluid interfaces.
- No credible human clinical pathway exists without reproducible large-animal evidence of integrated sensorimotor, respiratory, autonomic, pain, and long-term survival outcomes.
- Research should remain component-based and governed by strict ethical gates rather than calendar-based promises of a first human operation.

1. Introduction

Transplantation replaces a failing biological component with a viable donor cell, tissue, or organ. The brain, however, is not merely another perfused organ: it is the substrate of consciousness, memory, agency, communication, and the neural control of breathing, circulation, endocrine regulation, movement, sensation, and behavior. Consequently, a proposed brain transplant raises simultaneous surgical, immunological, neurological, legal, and philosophical questions that do not arise in ordinary solid-organ transplantation. WHO frameworks emphasize that transplantation must be therapeutically justified, ethically governed, voluntarily consented, and protected from exploitation [16,17].

The phrase “brain transplant” is frequently used imprecisely. An isolated whole-brain allotransplant would require removal of a donor brain and implantation into another cranium or body, with reconstruction of cerebral arteries and veins, brainstem-spinal continuity, cranial nerves, meninges, venous sinuses, cerebrospinal-fluid pathways, and hypothalamic-pituitary connections. By contrast, a cephalosomatic or whole-body transplant would preserve the recipient brain, skull, sensory organs, cranial nerves, and much of the upper airway, while attaching the head to a donor body. The latter reduces some cranial interfaces but still requires complete cervical spinal-cord reconnection and durable integration of vascular, respiratory, autonomic, musculoskeletal, immune, and endocrine systems [1-3].

This distinction matters because the anticipated clinical objective is also different. Transplanting a diseased brain would not cure a primary neurodegenerative disorder. A whole-body transplant would instead be conceived for a person whose brain is relatively preserved but whose body is catastrophically diseased. Such a proposal must therefore be compared with less radical alternatives, including multiorgan transplantation, mechanical support, gene and cell therapies, prosthetics, and palliative care. A high-risk intervention cannot be ethically justified merely because it is technically imaginable.

2. Scope, Definitions, and Review Method

2.1 Operational definitions

Table 1. Terminology used in this review

Term	Operational definition	Present status
Whole-brain allotransplantation	Removal of an intact donor brain and implantation into a recipient cranial/body system, requiring reconstruction of vascular, neural, meningeal, CSF, endocrine, and immune interfaces.	No validated large-animal or human procedure.
Cephalosomatic or whole-body transplantation	Attachment of a recipient head/brain to a donor body. Cranial nerves are largely preserved; the cervical spinal cord and neck structures are transected and must be reconstructed.	Historical animal feasibility for short-term perfusion only; no functional spinal reconnection or human procedure.
Neural cell transplantation	Implantation of neural stem/progenitor cells or lineage-specific neurons into the CNS.	Active preclinical and early clinical field; not equivalent to whole-brain transplantation.

Term	Operational definition	Present status
Ex vivo brain perfusion	Extracorporeal circulation of an isolated brain with oxygenated/cytoprotective perfusate.	Demonstrated selected cellular and metabolic recovery in porcine research.
Integrated neurological recovery	Coherent restoration of consciousness-relevant activity plus motor, sensory, respiratory, autonomic, endocrine, and behavioral function.	Not demonstrated after brain or whole-body transplantation.

2.2 Review approach

This article is a critical narrative review and research agenda rather than a systematic review or meta-analysis. Literature was assessed through 13 July 2026 using PubMed-indexed articles, major peer-reviewed journals, international consensus guidelines, WHO transplantation materials, and ISSCR guidance. Priority was given to original large-animal perfusion studies, historical primary surgical reports, contemporary spinal-cord repair literature, transplantation immunology, death-determination guidelines, and peer-reviewed ethical analyses. Search concepts included "brain transplantation," "cephalic exchange," "whole-body transplantation," "brain perfusion," "BrainEx," "OrganEx," "spinal cord regeneration," "vascularized composite allotransplantation," "brain death," "personal identity," and "stem-cell clinical translation." Because the field has inconsistent terminology and no clinical trials of whole-brain transplantation, the evidence was synthesized by component rather than pooled quantitatively.

Claims were classified as established, experimentally supported, plausible but unproven, or speculative. A central methodological rule was maintained throughout: preservation of cells or local synaptic activity was not interpreted as recovery of integrated brain function, and short-term vascular survival was not interpreted as functional transplantation.

3. Historical Evidence and What It Actually Demonstrated

3.1 Early cephalic-exchange experiments

In 1971, White and colleagues reported cephalic exchange transplantation in monkeys [1]. The experiments were historically important because they addressed cerebral protection, rapid vascular connection, and short-term perfusion of the transplanted cephalon. However, the cervical spinal cord was not functionally reconnected. The animals therefore could not achieve voluntary motor control of the donor body, normal sensation below the transection, or independent integrated respiratory function. Later historical reviews emphasized the vascular and preservation lessons of these models, not a completed neurological transplant [2,3].

These experiments are sometimes described in popular accounts as proof that head transplantation "worked." That description is scientifically misleading. A perfused, awake-appearing cephalon with preserved cranial responses is not equivalent to a neurologically integrated organism. The result demonstrated a limited form of temporary cephalic survival under intensive support, while simultaneously revealing the central failure point: the severed spinal cord.

3.2 BrainEx and OrganEx

BrainEx was an extracorporeal pulsatile-perfusion platform applied to isolated pig brains four hours after death. It restored microcirculation and selected molecular, metabolic, vascular, glial, and spontaneous synaptic functions while preserving aspects of cytoarchitecture. Crucially, the investigators reported no global electrocorticographic activity of the kind expected for organized whole-brain function [4]. BrainEx therefore challenged simple assumptions about the timing of cellular irreversibility, but it did not restore consciousness, revive an animal, or demonstrate that an isolated brain could be transplanted.

OrganEx adapted related perfusion concepts to the porcine whole body after one hour of warm ischaemia. It improved perfusion, tissue integrity, and selected molecular and cellular processes in multiple organs compared with conventional extracorporeal circulation controls [5]. Again, the study addressed cellular recovery and preservation biology. It did not demonstrate normal integrated organ function, recovery of consciousness, or a pathway to transplanting the brain. These experiments are most relevant as evidence that ischaemia-reperfusion injury may be modifiable, not that death or personhood can be reversed.

Table 2. Evidence hierarchy relevant to brain transplantation

Evidence domain	What has been demonstrated	What has not been demonstrated	Interpretive caution
Historical animal cephalic exchange	Short-term vascular perfusion and survival of a transplanted cephalon in primates.	Functional spinal-cord reconnection, durable autonomous survival, or human feasibility.	Vascular survival is not neurological integration.
BrainEx	Restoration/preservation of microcirculation and selected cellular, metabolic, glial, and synaptic functions in isolated pig brains.	Global organized electrical activity, consciousness, memory, or transplantability.	Cellular recovery is not revival of the person.
OrganEx	Improved cellular and tissue integrity in multiple pig organs after warm ischaemia.	Normal whole-body function or human organ-transplant protocol for the brain.	Molecular repair markers are surrogate outcomes.
Spinal-cord regeneration research	Axonal growth, remyelination, relay circuits, biomaterials, cells, and stimulation can improve selected outcomes in models and some patients.	High-bandwidth reconnection after complete cervical transection with full autonomic and sensorimotor integration.	Partial recovery after injury is not equivalent to surgical cord fusion.

Evidence domain	What has been demonstrated	What has not been demonstrated	Interpretive caution
Vascularized composite allotransplantation	Complex tissues such as face and limb can survive with microsurgery and immunosuppression.	Tolerance of a brain/head-body construct or elimination of chronic rejection.	Greater tissue complexity increases immune and rehabilitation burden.

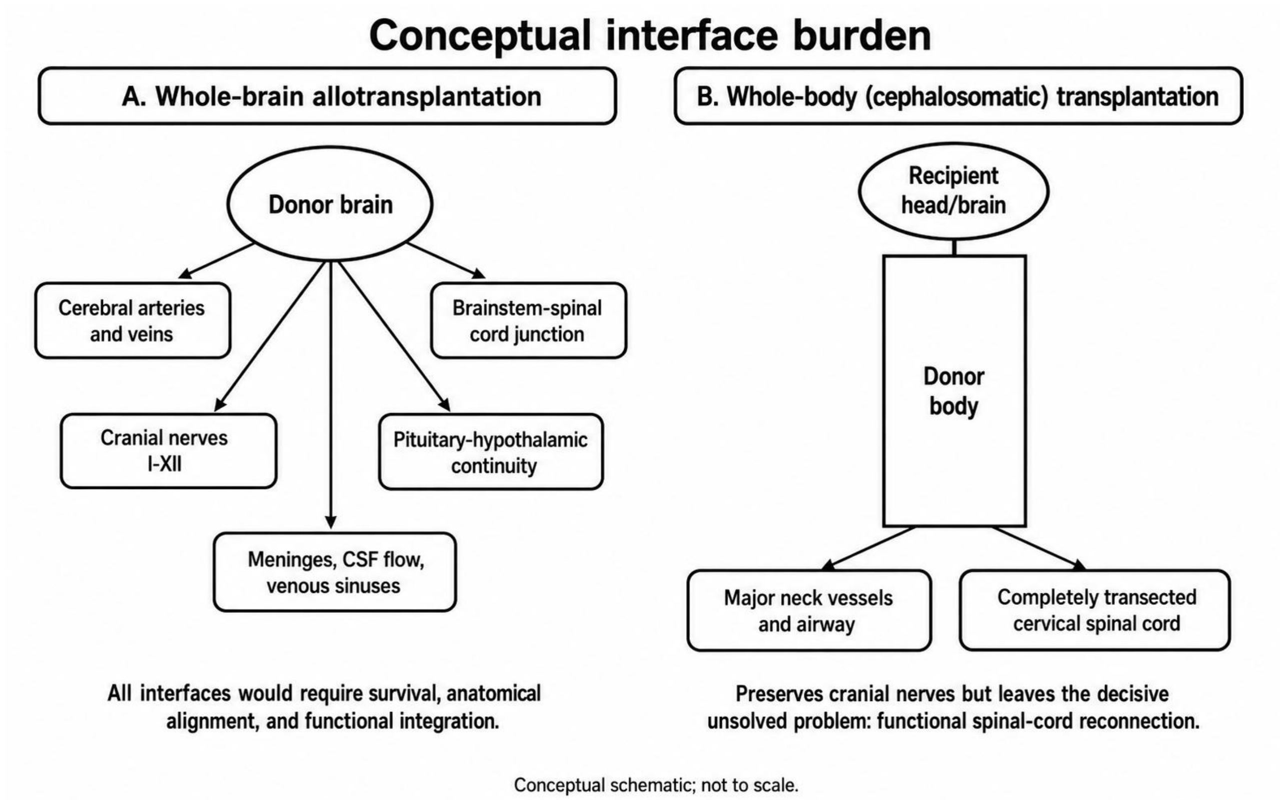

Figure 1. Conceptual anatomical interface burden. Whole-brain allotransplantation adds multiple neurovascular, cranial-nerve, meningeal, CSF, endocrine, and brainstem-spinal interfaces. Whole-body transplantation preserves most cranial interfaces but still requires complete cervical spinal-cord reconnection. Original schematic by the author; not to scale.

4. Scientific Feasibility by System

4.1 Cerebral preservation and ischaemia-reperfusion injury

The brain has high metabolic demand, limited energy reserves, and marked vulnerability to interruption of oxygen and glucose delivery. A transplant procedure would need to minimize warm ischaemia, avoid embolism, preserve the microvasculature and blood-brain barrier, control oedema, maintain temperature and acid-base balance, and re-establish cerebral autoregulation. BrainEx and OrganEx suggest that cytoprotective perfusion can extend the window for selected cellular recovery [4,5], but a clinically useful transplant would require much more: stable perfusion of every critical region, absence of diffuse microinfarction, preserved network dynamics, and long-term cognitive and behavioral function.

The relevant endpoint cannot be simple blood flow. It must include multimodal monitoring such as vascular imaging, oxygen extraction, intracranial pressure, cerebral metabolism, electroencephalography, evoked potentials, structural and diffusion MRI, histopathology, and longitudinal behavior. A perfused brain that cannot support integrated function would not meet a therapeutic threshold.

4.2 Vascular reconstruction

At minimum, a cephalosomatic transplant requires rapid anastomosis of major neck arteries and veins while preventing thrombosis, air embolism, venous congestion, and excessive reperfusion pressure. An isolated brain transplant is more difficult because the intracranial arterial inflow, venous sinuses, meningeal vessels, and skull-base relationships must be reconstructed. Even technically patent vessels may not guarantee capillary-level reperfusion. Endothelial injury can trigger oedema, inflammation, coagulopathy, and blood-brain-barrier disruption.

Modern microsurgery and vascularized composite allotransplantation show that complex vascular anastomoses are possible, but the tolerable ischaemia time and functional stakes are fundamentally different for the brain. The vascular problem is necessary but not sufficient: success at this level would only create the conditions in which neural integration could be tested.

4.3 Spinal-cord reconnection: the decisive barrier

A complete cervical transection interrupts descending motor pathways, ascending sensory pathways, propriospinal networks, sympathetic and parasympathetic control, respiratory drive to spinal motor neurons, and numerous modulatory systems. Adult mammalian central axons show limited spontaneous regeneration because of intrinsic growth constraints, inhibitory extracellular signals, inflammation, cavitation, glial scarring, loss of myelin, and failure to re-establish appropriate targets [6-10]. Contemporary research increasingly aims to create relay circuits, guide axonal growth, transplant neural progenitors, modulate the injury environment, and use electrical stimulation or brain-spine interfaces to exploit spared pathways. These approaches are scientifically important but do not amount to "fusing" two completely transected cervical cords.

For a transplant, the requirement is substantially more stringent than improvement after ordinary spinal-cord injury. Millions of axonal relationships need not be restored one-to-one, but enough correctly organized pathways must cross the interface to support stable breathing, posture, voluntary movement, sensation, continence, cardiovascular control, thermoregulation, sexual function, and protection from autonomic dysreflexia and neuropathic pain. Reconnection that produces uncontrolled motor activity, pain, or autonomic instability would not be a successful outcome.

A credible preclinical claim would require blinded, independently replicated evidence in large animals after anatomically verified complete cervical transection. Endpoints should include tract-specific electrophysiology across the lesion, independent ventilation, purposeful motor behavior, sensory discrimination, cardiovascular and bladder control, pain assessment, histological continuity, and survival measured in months or years. No published evidence currently satisfies this standard.

4.4 Cranial nerves, brainstem, and sensory systems

Whole-body transplantation preserves the recipient eyes, ears, olfactory structures, facial musculature, and cranial nerves, which is one reason it is technically less complex than isolated whole-brain transplantation. A true brain transplant would sever or disrupt the optic nerves, olfactory pathways, ocular motor nerves, trigeminal and facial systems, vestibulocochlear nerves, lower cranial nerves, and skull-base vasculature. Reconnecting these structures with high-fidelity function is beyond current clinical neurorepair. Loss of vision, swallowing, airway protection, facial sensation, hearing, balance, and communication would be likely even if cerebral perfusion were preserved.

4.5 Cerebrospinal fluid, meninges, venous sinuses, and intracranial pressure

The brain operates within a closed cranial compartment. A transplanted brain would require watertight reconstruction of dura and arachnoid, controlled cerebrospinal-fluid production and drainage, venous-sinus outflow, and stable intracranial compliance. Mismatch between brain volume and skull, venous congestion, hydrocephalus, CSF leak, infection, and intracranial hypertension could be rapidly fatal. These interfaces are often omitted from simplified transplant diagrams but are essential to viability.

4.6 Hypothalamic, pituitary, endocrine, and autonomic integration

The hypothalamus coordinates temperature, thirst, appetite, circadian rhythms, stress responses, reproductive endocrinology, and autonomic function. An isolated brain transplant would have to preserve or reconstruct the hypothalamic-pituitary axis and its specialized portal circulation. A head/body transplant would retain the recipient hypothalamus and pituitary but connect them to donor peripheral endocrine organs. Hormonal set points, receptor sensitivity, autonomic reflexes, and immune-metabolic interactions may not immediately match. Intensive endocrine monitoring and replacement would be necessary, yet replacement hormones cannot substitute for complete neural-autonomic integration.

4.7 Immunology and graft rejection

The central nervous system is not absolutely isolated from immunity. Brain borders, meninges, endothelial cells, perivascular macrophages, microglia, lymphatic drainage, and circulating immune cells participate in surveillance and inflammation [11,12]. A whole-brain graft would carry donor endothelium, microglia, meninges, and potentially other immunogenic tissues. A whole-body transplant would resemble an extreme vascularized composite allograft, exposing the recipient to a large donor antigen burden. Acute rejection, chronic vasculopathy, donor-specific antibodies, infection, malignancy, nephrotoxicity, and metabolic complications of immunosuppression would be foreseeable [13,14].

Immune tolerance would be preferable to lifelong high-dose immunosuppression, but reliable tolerance for complex allografts remains investigational. Neurological decline from rejection might be difficult to distinguish from ischaemic injury, infection, drug toxicity, or spinal-interface failure. Unlike a kidney or limb graft, failure might not permit explantation or return to the pre-transplant state.

4.8 Infection, malignancy, and rehabilitation

The procedure would combine prolonged surgery, massive transfusion, invasive monitoring, open neural tissue, airway and gastrointestinal contamination risks, indwelling devices, and immunosuppression. The likelihood of bacterial, fungal, viral, and opportunistic infection would be high. Long-term rehabilitation would exceed that required for face, limb, or ordinary spinal transplantation research because the patient would need to learn control of a donor body while managing profound sensory, autonomic, and psychological disruption. Rehabilitation itself would be an experimental intervention requiring objective endpoints and independent monitoring.

Table 3. Major barriers and current readiness

Barrier	Minimum requirement for feasibility	Current evidence	Readiness judgment
Cerebral preservation	Reversible control of ischaemia, oedema, BBB injury, and microvascular failure with preserved network function.	Selected cellular recovery in porcine perfusion models.	Early component evidence; no transplant-level validation.

Barrier	Minimum requirement for feasibility	Current evidence	Readiness judgment
Large-vessel and microvascular perfusion	Rapid anastomosis plus uniform capillary reperfusion and autoregulation.	Microsurgical capability exists; brain-specific integrated evidence absent.	Technically plausible component, biologically unvalidated.
Complete cervical cord reconnection	Durable cross-lesion motor, sensory, respiratory, autonomic, and pain-safe conduction.	Partial regeneration/neuromodulation in models and selected patients with injury.	Decisive unsolved barrier.
Cranial-nerve reconstruction	Functional vision, eye movement, hearing, swallowing, facial control, and airway protection.	Peripheral nerve repair is possible; high-fidelity multi-cranial reconstruction is not.	Not feasible for isolated whole-brain transplant.
CSF/meningeal/venous integration	Watertight closure, normal pressure, drainage, and venous outflow.	Neurosurgical techniques address individual elements.	Complex but secondary to neural integration.

Barrier	Minimum requirement for feasibility	Current evidence	Readiness judgment
Immune control	Long-term graft survival without unacceptable infection, malignancy, or organ toxicity.	VCA experience shows both feasibility and chronic rejection burden.	Insufficient for a non-rescuable brain/head-body graft.
Ethical/legal governance	Valid consent, identity and death rules, donor protection, justice, welfare, and stopping criteria.	General frameworks exist; procedure-specific consensus absent.	No acceptable clinical governance pathway.

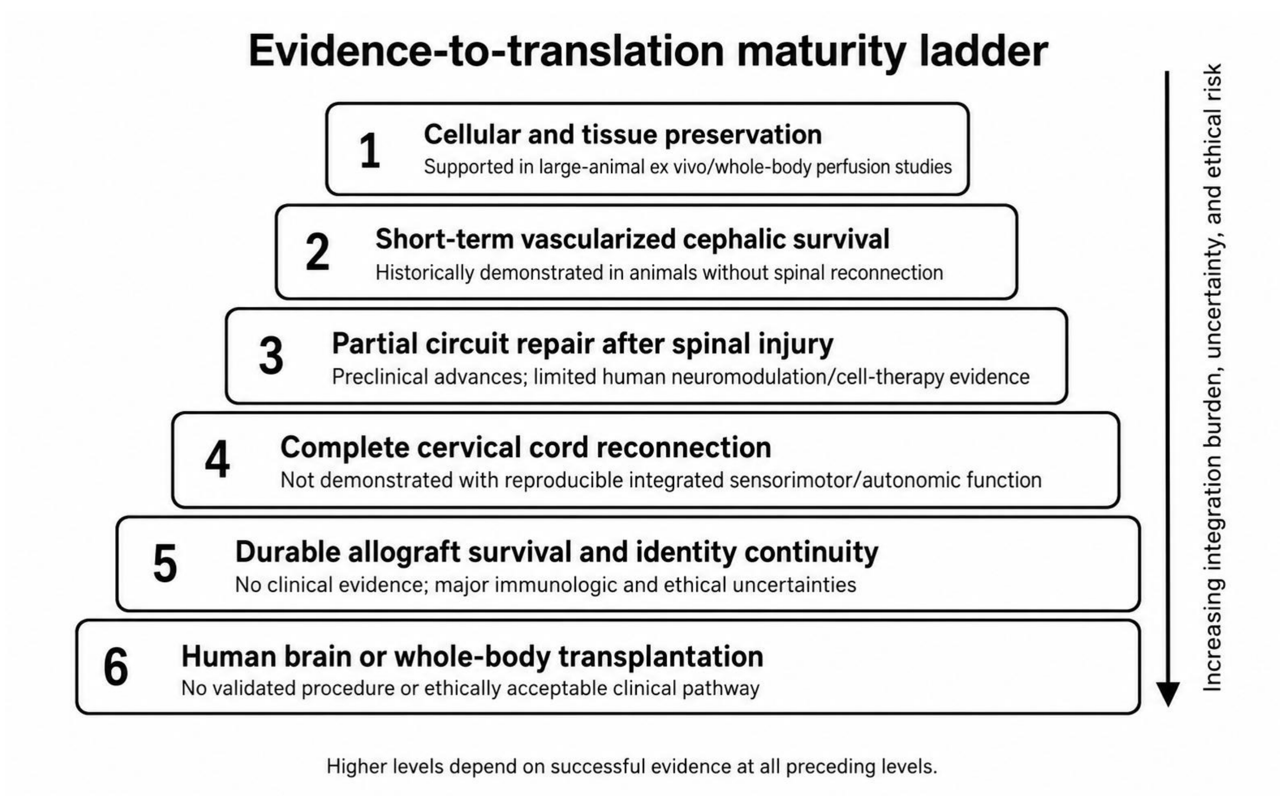

Figure 2. Evidence-to-translation maturity ladder. Existing evidence is concentrated at the levels of cellular preservation, short-term perfusion, and partial neural repair. Each higher level requires qualitatively new evidence rather than extrapolation from lower levels. Original schematic by the author.

5. Potential Enabling Technologies - and Their Limits

5.1 Hypothermia and machine perfusion

Hypothermia reduces metabolic demand, while normothermic or subnormothermic perfusion may permit physiological assessment and targeted treatment. Future systems could combine oxygen carriers, anti-oedema agents, free-radical control, anticoagulation, metabolic substrates, inflammatory modulation, and real-time imaging. However, preservation technologies cannot solve severed neural pathways. Their role is to protect viable tissue long enough for other repairs to be attempted.

5.2 Biomaterials, cells, gene regulation, and engineered relay circuits

Hydrogels, aligned scaffolds, extracellular-matrix materials, Schwann cells, oligodendrocyte progenitors, neural stem cells, organoid-derived tissue, and gene-regulatory approaches are being studied to support axonal growth, remyelination, and circuit reconstruction [6-10,15]. A rational strategy may eventually combine structural guidance, growth-permissive molecular states, cell replacement, and activity-dependent rehabilitation. The risk of ectopic growth, tumor formation, maladaptive connectivity, seizures, pain, and immune reactions must be treated as central, not peripheral, endpoints.

5.3 Neuroprosthetics and brain-spine interfaces

Neural interfaces can decode cortical intention and stimulate spinal circuits below an injury. They may restore selected functions without requiring full biological regrowth and are therefore a more realistic near-term strategy for paralysis. Yet they depend on implant durability, calibration, power, signal stability, and the existence of usable downstream circuits. A device-mediated bridge might supplement incomplete biological reconnection, but it would not by itself restore endocrine, sensory, autonomic, or immune integration after whole-body transplantation.

5.4 Immune tolerance and localized immunosuppression

Cellular therapies, regulatory immune cells, mixed chimerism, and localized drug-delivery systems are being investigated to reduce rejection and systemic toxicity [13,14]. For a brain/head-body graft, tolerance would need to be robust, durable, and compatible with infection control. Because graft failure would be catastrophic and potentially irreversible, the acceptable uncertainty is much lower than for experimental limb or face transplantation.

6. Ethical, Legal, and Social Analysis

6.1 Identity and the "recipient" of the transplant

Ordinary transplantation assumes that the recipient remains the same legal and moral person after receiving an organ. A whole-body transplant reverses this intuition: the person associated with the brain and head would ordinarily be regarded as the continuing individual, while the donor body would be a graft. A whole-brain transplant creates an even sharper question because the donor brain would likely carry memories, personality, and agency, while the recipient body and legal identity would belong to another person. Ethical analyses of whole-body transplantation therefore focus on personal identity, family relations, parentage, criminal and civil responsibility, ownership, and the status of the donor body [18,19].

The paper adopts a practical neuroethical position: continuity of the functioning brain is strong evidence for continuity of the person, but this does not eliminate uncertainty about severe cognitive injury, memory loss, delirium, dissociation, or altered agency. Consent must address not only mortality and paralysis but also the possibility of survival with profound psychological and identity disruption.

6.2 Donor consent and death determination

Donation must occur only after valid death determination under applicable law and accepted medical standards. Contemporary consensus guidelines define death by neurologic criteria through permanent cessation of brain function and require rigorous exclusion of confounders [20,21]. Brain-preservation research complicates public understanding because cellular activity can persist or be restored without recovery of the integrated functions that determine consciousness or personhood. Programs must therefore communicate the difference between cellular viability and reversal of death with exceptional precision.

A donor body for whole-body transplantation would require explicit authorization for an unprecedented use with major implications for appearance, reproduction, family recognition, and public exposure. Standard organ-donor consent should not be presumed to cover such a procedure. Commercialization, coercion, vulnerable populations, and organ trafficking are unacceptable under WHO principles [16,17].

6.3 Risk-benefit proportionality and therapeutic misconception

A first-in-human intervention would predictably involve death, paralysis, pain, infection, rejection, cognitive injury, and prolonged intensive care. The prospective benefit is hypothetical. Under these conditions, informed consent alone cannot make a trial ethical; scientific validity and a favorable risk-benefit ratio are independent requirements. Publicity-driven promises may intensify therapeutic misconception, especially among people with severe disability. ISSCR guidance emphasizes rigorous preclinical justification, independent review, transparent communication, and accurate representation of uncertainty [22].

6.4 Disability rights and social meaning

Framing disability as a fate worse than death can stigmatize people living with paralysis or multisystem disease. Research priorities should be shaped with disabled persons, rehabilitation specialists, caregivers, and patient advocates. A speculative transplant should not divert resources from accessible care, prevention, assistive technology, pain treatment, or evidence-based rehabilitation. The goal of research must be expanded capability and well-being, not conformity to a narrow model of bodily normality.

6.5 Animal welfare

Historical transplant experiments caused profound suffering and would not automatically satisfy modern standards. Any future animal work must use the least sentient model capable of answering the question, avoid redundant demonstrations, include analgesia and humane endpoints, and undergo specialized scientific and ethics review. Experiments capable of generating organized activity in isolated brains or human-neural chimeric systems require additional oversight because ordinary welfare indicators may be inadequate [22].

Table 4. Ethical and governance requirements before any escalation

Domain	Required safeguard	Reason for requirement
Scientific validity	Independent replication, preregistered endpoints, complete reporting of failures, and no reliance on promotional claims.	High-risk trials are unethical when the evidence base is weak or selectively reported.
Consent	Procedure-specific donor and recipient consent, capacity assessment, psychological evaluation, and independent advocates.	Identity, reproductive, family, and catastrophic-disability risks exceed ordinary transplant consent.
Death determination	Strict separation between treating, death-determination, procurement, and research teams.	Protects public trust and prevents conflict of interest.
Animal welfare	Specialized review, analgesia, humane endpoints, monitoring for consciousness-compatible activity, and stop rules.	Conventional endpoints may not capture suffering in isolated-brain or cephalic models.
Justice	Transparent selection criteria, prohibition of payment for bodies/organs, and assessment of opportunity cost.	Prevents exploitation and inequitable allocation.
Communication	Balanced public statements that distinguish cellular activity, consciousness, and clinical feasibility.	Reduces hype, therapeutic misconception, and misinformation.

7. A Responsible Research Roadmap

A responsible roadmap should be milestone-based rather than date-based. It should not begin with planning a human operation. It should begin with component problems that have independent medical value, such as brain preservation after cardiac arrest, spinal-cord repair, immune tolerance, and neuroprosthetics. Each phase should have predefined go/no-go criteria and should be subject to external replication.

Table 5. Staged roadmap with go/no-go criteria

Phase	Research objective	Minimum evidence to proceed	Stop/redirection criteria
0. Governance and definitions	Standardize terminology, endpoints, death criteria, welfare monitoring, and conflict-of-interest rules.	International multidisciplinary consensus and public/patient engagement.	No agreed definitions, unclear legal authority, or promotional pressure overriding science.
1. Neurovascular preservation	Improve ex vivo and in situ preservation while preventing oedema and unrecognized organized activity.	Reproducible large-animal microvascular and network preservation with validated monitoring.	Evidence of suffering risk without adequate detection/termination; irreproducible results.
2. Spinal-cord reconstruction	Restore conduction and function after verified complete cervical transection.	Independent replication of respiratory, autonomic, motor, sensory, and pain-safe outcomes with long survival.	Only partial reflexes, uncontrolled pain, autonomic instability, or dependence on spared pathways.

Phase	Research objective	Minimum evidence to proceed	Stop/redirection criteria
3. Integrated large-animal physiology	Test compatibility of preservation, neural repair, immune control, and rehabilitation without full transplantation where possible.	Stable multidomain function and acceptable welfare over clinically meaningful duration.	High mortality, severe distress, rejection, infection, or inability to define benefit.
4. Regulatory consideration	Evaluate whether any human protocol could meet scientific and ethical standards.	Compelling superiority over alternatives, durable animal evidence, independent regulatory and ethics approval.	Unfavorable risk-benefit, inadequate consent framework, inequity, or non-rescuable failure modes.

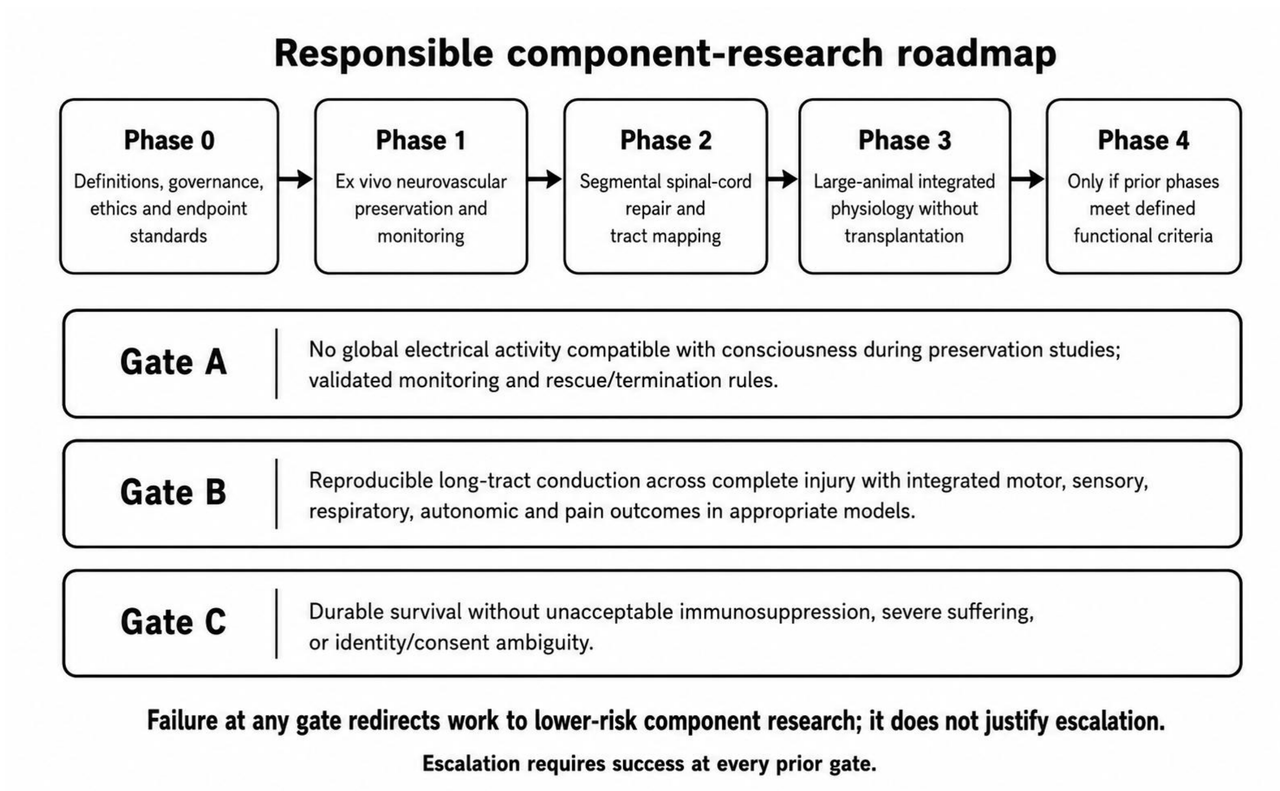

Figure 3. Responsible component-research roadmap. Progression depends on functional and ethical gates, not predictions of a calendar date for human transplantation. Original schematic by the author.

8. Proposed Minimum Outcome Set for Preclinical Claims

Because vague claims of "survival" or "movement" are easily misinterpreted, any future preclinical report should publish a minimum multidomain outcome set:

- Anatomical verification of the lesion and repair interface, including evidence that outcomes are not explained by spared pathways.
- Continuous cerebral perfusion, oxygenation, intracranial pressure, EEG, and evoked-potential monitoring with prespecified thresholds.
- Independent ventilation and airway protection, not merely machine-supported gas exchange.
- Purposeful motor behavior, postural control, and task performance assessed by blinded observers.
- Sensory discrimination and pain-related outcomes, including evidence against severe neuropathic pain or dysesthesia.
- Autonomic outcomes: blood-pressure stability, heart-rate control, thermoregulation, bowel and bladder function, and absence of recurrent autonomic crises.
- Cognition, memory, communication, affect, sleep-wake organization, and behavior when the model permits meaningful assessment.
- Immune, infectious, endocrine, metabolic, and histopathological outcomes over a duration sufficient to detect chronic failure.
- Complete accounting of deaths, exclusions, adverse events, protocol deviations, and negative experiments.

9. Clinical Indications and Alternatives

A procedure cannot be evaluated without a defined clinical indication. Primary brain diseases are generally poor indications because moving the diseased brain does not remove the pathology. A whole-body transplant has been proposed for preserved cognition with terminal systemic disease, but this population is heterogeneous and often medically fragile.

Alternative strategies may include transplantation of one or more failing organs, ventricular assist or respiratory support, gene therapy, targeted cell transplantation, rehabilitation, neural interfaces, or symptom-focused care. An ethically acceptable study would need to show that these alternatives cannot provide comparable benefit with less risk.

Reproductive and genetic questions also differ from ordinary transplantation. Gametes produced by the donor body would be genetically related to the body donor, not necessarily to the brain/head recipient. Pregnancy, fertility, parentage, and consent for reproductive use would require specific law and policy. These issues are not secondary; they are foreseeable consequences of the intervention.

10. Limitations of This Review

This is a narrative synthesis of a sparse and heterogeneous evidence base. Historical reports used methods and welfare standards that differ from current practice. Modern perfusion studies were not transplantation studies, and spinal-cord research spans different injury types, species, outcome measures, and degrees of pathway sparing. No meta-analysis can bridge these categories. The field also contains high-profile claims that have not been independently reproduced. Accordingly, the roadmap offered here is a framework for evaluating evidence, not a prediction that whole-brain or whole-body transplantation will become feasible.

11. Conclusion

Brain transplantation is not a current medical procedure and should not be represented as one. Historical primate experiments established limited short-term cephalic perfusion without spinal-cord reconnection. BrainEx and OrganEx established that selected cellular and molecular processes can recover after prolonged ischaemia under experimental conditions, but they did not restore consciousness or demonstrate transplantability. Contemporary spinal-cord regeneration, biomaterials, cell therapy, and neuroprosthetic research are progressing, yet none has achieved the reproducible, high-bandwidth, pain-safe, autonomically integrated reconnection required after complete cervical transection.

The scientifically defensible position is therefore neither categorical metaphysical impossibility nor near-term clinical optimism. It is disciplined uncertainty. Component technologies may produce major benefits for cardiac arrest, stroke, spinal-cord injury, neurodegeneration, organ preservation, and rehabilitation even if a whole-brain or whole-body transplant is never justified. Human experimentation should not be considered unless independent large-animal evidence demonstrates durable integrated function, immune control, acceptable welfare, and superiority to less harmful alternatives. Until then, the responsible research program is preservation, repair, tolerance, transparent reporting, and neuroethical governance - not a first human transplant attempt.

Declarations

Author contributions

SAMUELSON G conceived the review, defined its scope, synthesized the literature, developed the figures and tables, and prepared the manuscript.

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Competing interests

The author declares no competing financial or non-financial interests.

Ethics approval and consent to participate

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Consent for publication

Not applicable. The manuscript contains no identifiable personal data.

Data availability

No new datasets were generated or analyzed. All sources are cited in the reference list.

Clinical trial registration

Not applicable. This manuscript does not report a clinical trial.

Disclaimer

This manuscript is an academic critical review, not medical advice, a surgical protocol, or authorization to conduct experiments. Any research involving neural tissue, transplantation, animals, or human participants requires competent institutional, regulatory, biosafety, and ethics oversight.

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